सी एस आई आर - राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
डॉ. के. एस. कृष्णन् मार्ग, नई दिल्ली-110012, भारत
Dr. K. S. Krishnan Marg, New Delhi-110012 INDIA
दूरभाष/Phone : +91-11-4560, 8441, 8610, 9447, फैक्स/Fax : 91-11-4560 8448
ई-मेल/E-mail : cfct@nplindia.org, वेबसाईट/Website : www.nplindia.org

अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.
25060295/D1.01/C-050

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
18.08.2025	18.08.2028	1	2

1. Calibrated for : Wika Process Solutions India Pvt. Ltd.
H.B. No. 40, Revenue Estate, Prithla - Dhatir Road
Village - Dudhola , Palwal
Haryana - 121 102, India

Customer's Ref. No.: Letter Dated : 09.05.2025

2. Description and identification of instrument

500 kg to 200 kg (15 Nos.) : Rectangular parallelepiped cast iron weights with hook having adjustment cavity at the side.(for 500 kg & 200 kg)

50 kg (2 Nos.) : Hexagonal cast iron weights with hook having adjustment cavity at the bottom.(for 50 kg)

Assumed density (d) : (7 700 ± 400) kg/m^3; k = 2

3. Environmental conditions : Ambient conditions (for 500 kg & 200 kg)
: Temperature : (23.0 ± 1.5) °C } (for 50 kg)
Relative humidity : (50 ± 10) %

4. Standard(s) used with associated Measurement uncertainty : NPL working standard (Sub-Div.#1.01/M/20kg/00 & Sub-Div.#1.01/M/500 kg/01) of mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. Metrological Traceability of Standard(s) used : The standard(s) used for calibration is(are) traceable to the National Standard which realize the units of quantities according to the International System of Units (SI).

6. Principle/Methodology of Calibration & Calibration Procedure number : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/CP#WT/M-02. Method of comparison with the CSIR-NPL working standard(s) using substitution weighing. The reported mass value(s) is(are) the conventional mass value(s) (M_c) related to the true mass value(s) (M_t by formula : $M_c = M_t \times [1 - 1.2 \times (1/d - 1/8\,000)]$.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by: श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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सीएसआईआर CSIR भारत का नवाचार इंजन The Innovation Engine of India

अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.
25060295/D1.01/C-050

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
18.08.2025	18.08.2028	2	2

7. Results :

Denomination	Mass Value (kg)	Uncertainty (kg)
500 kg - 1	500.005 4	± 0.008 0
500 kg - 2	500.004 2	± 0.008 0
500 kg - 3	500.005 0	± 0.008 0
500 kg - 4	500.005 0	± 0.008 0
500 kg - 5	500.005 0	± 0.008 0
500 kg - 6	500.005 0	± 0.008 0
500 kg - 7	500.005 0	± 0.008 0
500 kg - 8	500.005 0	± 0.008 0
500 kg - 9	500.004 0	± 0.008 0
500 kg - 10	500.003 0	± 0.008 0
500 kg - 11	500.004 0	± 0.008 0
500 kg - 12	500.003 0	± 0.008 0
500 kg - 13	500.002 0	± 0.008 0
200 kg - 1	200.000 5	± 0.003 0
200 kg - 2	200.000 1	± 0.003 0
50 kg - 1	50.000 0	± 0.000 8
50 kg - 2	50.000 0	± 0.000 8

The reported uncertainty is at coverage factor $k = 2$ which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of Calibration : (04-06).08.2025

9. Remarks : Mass value(s) of the weight(s) is(are) within the maximum errors permissible in M_1 accuracy class of weights as per OIML R 111-1 : 2004.

अंशांकनकर्ता:
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प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by:
National Physical Laboratory Director's Nominee
डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam